CONFIDENTIAL

UNIVERSITI TEKNOLOGI MALAYSIA
FACULTY OF COMPUTING

ANSWER SCHEME TEST 1

SEMESTER I 2022/ 2023

SUBJECT CODE : **SECJ1013**

SUBJECT NAME : **PROGRAMMING TECHNIQUE I**

YEAR/COURSE : **1 (SECJH/SECVH/SECBH/SECRH/SECPH)**

TIME : **8.00 PM – 10.00 PM (2 HOURS)**

DATE : **13th DECEMBER 2022 (TUESDAY)**

VENUE : **BK1 – BK7, N28**

INSTRUCTIONS TO THE STUDENTS:

This test book consists of two parts:

PART A:	4 QUESTIONS	(40 MARKS)
PART B:	6 QUESTIONS	(60 MARKS)
TOTAL		**(100 MARKS)**

ANSWER ALL QUESTIONS IN THIS BOOKLET IN THE SPACES PROVIDED.

Name	**ANSWER SCHEME**
Matric No.	
Section	
Lecturer's Name	

This question booklet consists of **16 pages** inclusive of the cover page.

PART A: PROBLEM-SOLVING QUESTIONS [40 MARKS]

QUESTION 1 (7 Marks)

```
1.  Start
2.  deposit = 0.1
3.  Get name, duration, package, custType
4.  If (package = 1) //1M
    4.1 rental = 150 //0.5M
5.  Else_If (package = 2) //0.5M @ Else
    5.1 rental = 250 //0.5M
6.  End_If
7.  If (custType = 'R') //1M
    7.1 disc = 0.25 //0.5M
8.  Else_If (custType = 'N') //0.5M @ Else
    8.1 disc = 0.00 //0.5M
9.  End_If
10. price = rental * duration
11. dep_payment = deposit * price
12. tot_discount = disc * price
13. balance = price - tot_discount - dep_payment //2M
14. Display name, dep_payment, tot_discount, balance
15. End
```

Note: The order of the tests is unimportant. It could be similar to the answer above or vice versa (test ***custType*** first, then test the **package**).

QUESTION 2 (8 Marks)

a) Pseudocode 1 (4 marks)

Table 1: Tracing table for Question 2(a)

i	Output
2	**0.2 //1M**
5	**0.5 //1M**
10	**25 //1M**
20	**2 //1M**

b) Pseudocode 2 (4 marks)

Table 2: Tracing table for Question 2(b)

unitsUsed	usageType	Output
300	Casual	**210 //1M**
150	Heavy	**90 //1M**
1020	Casual	**510 //1M**
0	Heavy	**Invalid Input //1M**

QUESTION 3 **(12 Marks)**

t	y	k	k < nit	Output
1	*1*	*0*	True	
3	-1	1	True	
5	-7	2	True	
7	-17	3	True	
9	-31	4	True	
11	-49	5	False	
	49			13 //1M

Calculation:

nit = 1 + (8 / 2) = 1 + 4 = 5 //1M

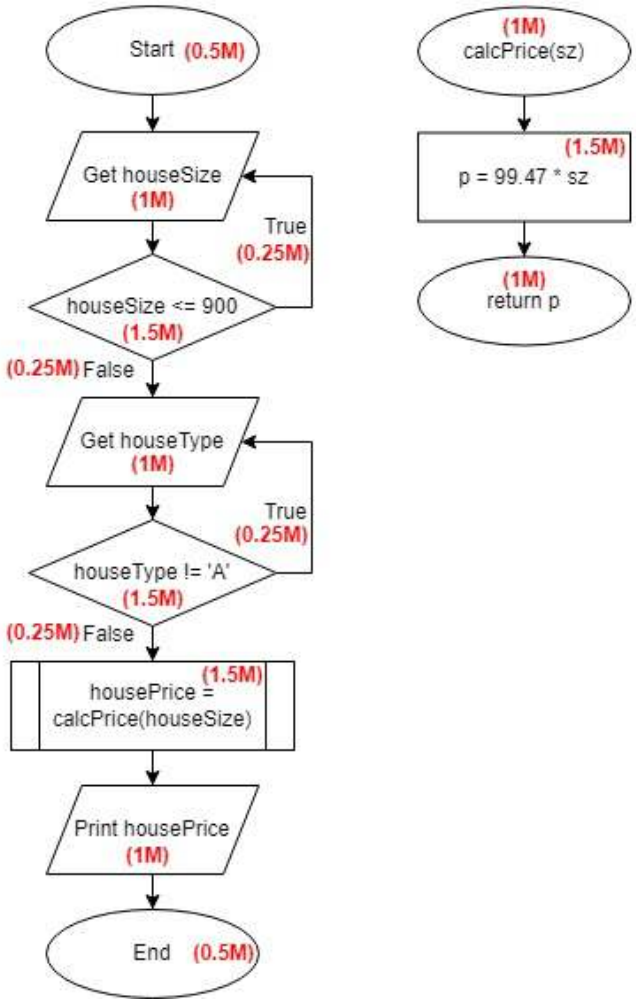
sum = 5 + 49 + 11 = 65 //1M

res = 65 / 5 = 13 //1M

0.25M each = 12 * 0.25 = 3M

0.5M each = 10 * 0.5 = 5M

QUESTION 4 **(13 Marks)**



PART B: PROGRAMMING QUESTIONS [60 MARKS]

QUESTION 1 (6 Marks)

```
int a = 8, b = 12, c = 25, d = 4, x;
float f = 4.5;
```

a) (3 marks)

```
x = a + b * (c++ / 31 % 13) - d;
x = 8 + 12 * (25 / 31 % 13) - 4; //0.5M
x = 8 + 12 * (0 % 13) - 4; //0.5M
x = 8 + 12 * 0 - 4; //0.5M
x = 8 + 0 - 4; //0.5M
x = 8 - 4; //0.5M
x = 4 //0.5M
```

b) (3 marks)

```
f -= b * ++a % static_cast<int>(f) + d;
f -= 12 * 9 % static_cast<int> 4.5 + 4; //0.5M
f -= 12 * 9 % 4 + 4; //0.5M
f -= 108 % 4 + 4; //0.5M
f -= 0 + 4; //0.5M
f -= 4 //0.5M
f = 4.5 - 4 //optional
f = 0.5 //0.5M
```

c) (2 marks)

```
d += 5 * 3 + (a++);
d += 5 * 3 + (8); //0.5M
d += 15 + 8; //0.5M
d += 23; //0.5M
d = 4 + 23; //optional
d = 27 //0.5M
```

d) (2 marks)

```
x = (d + ++b - f);
x = (4 + 13 - 4.5); //0.5M
x = (17 - 4.5); //0.5M
x = 12.5 //optional
x = 12 //1M
```

QUESTION 2 (5 MARKS)

Table 4: Program segment and output for Question 4

Line	Code	Output
1	`int h = 12, k = 7;`	
2	`bool m = false;`	
3		
4	`cout << ((h == 12) || (h > k)) << endl;`	**1**
5	`cout << ((h != 12) && (k = h)) << endl;`	**0**
6	`cout << ((h = 10) && (k = m)) << endl;`	**0**

7	`cout << ((h == 10) || !(h >= m)) << endl;`	**1**
8	`cout << ((h != 10) && (k != m)) << endl;`	**0**

QUESTION 3 **(12 Marks)**

a) Using **`switch-case`** statements (6 marks)

```
int main()
{
  int code;

  cout << "Enter code: ";
  cin >> code;

  switch(code) { //1m //0.25M
        case 1: cout << "Special"; break; //1.5m
        case 2: //0.5m
        case 4: cout << "Even"; break; //1.5m
        default: cout << "Awkward"; //1m
  } //0.25M
  //Note: Each cyan statement = 0.5m

  return 0;
}
```

b) Using **`if-else`** statements (6 marks)

```
int main()
{
  int n1, n2, n3, largest;

  cout << "Enter 3 numbers: ";
  cin >> n1 >> n2 >> n3;

  if (n1 > n2 && n1 > n3) //2.5m //0.25m
        largest = n1; //0.5m
  else if (n2 > n3) //1.5m @ else if (n2 > n1 && n2 > n3)
        largest = n2; //0.5m
  else //0.25m
        largest = n3; //0.5m
  //Note: Each cyan statement = 0.5m & each yellow statement = 1m

  cout << "Largest number: " << largest;
  return 0;
}
```

QUESTION 4 **(13 Marks)**

a) (7 marks)

Table 5: Answer for Question 4(a)

Line	Corrected Statements
2	`using namespace std; //missing using namespace std;` **//1m**
6	`for (int j = i; j > 0; j -= 5)` **//2m**
7	`if ((i + j) % 4 != 0) //remove ;` **//2m**
10	`cout << "j = " << --j //remove ; //Option 1` **//1m**
11	`cout << " i = " << i << endl; //Option 2` **//1m**
13	`i /= 2;` **//1m**

Note: Error 6 can occur on line 10 **or** line 11. Choose **only one**.

b) (6 marks)

```
j = 14 i = 25 //1m + 1m = 2m
j = 11 i = 12
j = 5 i = 6
```

Note: Each output for i and j = 1m

QUESTION 5 **(10 Marks)**

```
x: 1
y: 5
y: 6
x = 2
x = 3
z is 1
x is 4
1
0
0
```

Note: Each output (each line) = 1m

QUESTION 6 **(10 Marks)**

```
#include <iostream>
using namespace std;

int main()
{
    double speed;
    double hours;
    double distance;

    cout << "What is the speed of the vehicle in km/h? "; //0.5M
    cin  >> speed; //0.5M

    //Validate the speed is not negative.
    while (speed < 0) //1M
    {
       cout << "Please enter a non-negative number for speed: "; //0.5M
       cin  >> speed; //0.5M
    }

    //Get the hours traveled.
    cout << "For how many hours has it traveled? "; //0.5M
    cin  >> hours; //0.5M

    //Validate the hours is not less than 1.
    while (hours < 1) //1M
    {
       cout << "Please enter a 1 or greater for hours: "; //0.5M
       cin >> hours; //0.5M
    }

    //Display the table headings.
    cout << "\nHour    Distance Traveled in km \n";
    cout << "---------------------------------\n";

    //Display the distances traveled for each i.
    for (int i = 1; i <= hours; i++) //2M
    {
       distance = speed * i; //1M

       //Display the hours and distance.
       cout << i << "\t"  //0.5M
            << distance << endl; //0.5M
    }
    //Note: Each yellow statement = 0.5m

    return 0;
}
```